Entanglement Entropy Scaling as an Automatic Scope Gate: Validation on the Spin-$\frac{1}{2}$ XXZ Chain

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Abstract

We introduce and validate a *scope gate*: an automatic decision procedure that classifies a quantum many-body system as either amenable to holographic tensor-network methods (IN-SCOPE) or not (OUT-OF-SCOPE) based solely on the finite-size scaling of half-chain entanglement entropy $S(L)$. The criterion rests on Akaike's Information Criterion with correction (AICc) model selection between a logarithmic (Calabrese–Cardy) model and an exponentially-saturating model. When $\Delta\mathrm{AICc} \equiv \mathrm{AICc}_{\mathrm{sat}} - \mathrm{AICc}_{\mathrm{log}} \leq -2$ the saturating model is favoured and the system is classified IN-SCOPE; when $\Delta\mathrm{AICc} \geq +2$ the log model is favoured and the system is classified OUT-OF-SCOPE. We validate the gate on the $S = \frac{1}{2}$ XXZ spin chain, computing $S(L)$ by exact diagonalization (ED) at $L = 8, 12, 16$ and density-matrix renormalization group (DMRG) at $L = 20, \ldots, 44$ ($L \bmod 4 = 0$ throughout) for anisotropies $\Delta = 0.5$ and $\Delta = 2.0$. The ED and DMRG data agree to relative precision $< 10^{-8}$ at shared system sizes, confirming their poolability. For $\Delta = 0.5$ (critical XY phase) the log model is overwhelmingly preferred ($\Delta\mathrm{AICc} = +20.1$, OUT-OF-SCOPE), while for $\Delta = 2.0$ (gapped Néel phase) the saturating model is preferred ($\Delta\mathrm{AICc} = -21.1$, IN-SCOPE). Near the Kosterlitz–Thouless transition ($\Delta \approx 1$) the gate correctly returns INCONCLUSIVEbecause the exponentially large correlation length $\xi_{\mathrm{KT}} \sim e^{\pi/\sqrt{\Delta-1}}$ exceeds the accessible system sizes, an outcome that is physically correct rather than a failure. The AICc's finite-sample correction provides a natural safeguard: when data are too sparse to constrain the saturating model ($n \leq 4$ for a three-parameter fit), the penalty term diverges and the gate is forced to return INCONCLUSIVErather than a false positive. These results establish entanglement entropy scaling as a reliable, numerically inexpensive pre-screening tool for selecting appropriate computational methods.

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1 Introduction

Tensor-network methods such as matrix product states (MPS) and the multi-scale entanglement renormalization ansatz (MERA) achieve their efficiency through the assumption that entanglement in the target state is geometrically constrained—specifically, that the half-chain entanglement entropy S saturates to a finite constant as the system size grows [1, 2, 3]. For gapped one-dimensional systems the area law $S \sim \text{const}$ holds by the Hastings theorem [4], whereas critical systems with a conformal field theory (CFT) description obey the Calabrese–Cardy (CC) logarithm $S(L) = \frac{c}{3} \ln L + \text{const}$ for periodic boundaries, or $\frac{c}{6} \ln L$ for open boundaries [5].

A natural question is whether one can *automatically* decide, before investing in a full tensor-network simulation, whether a system will be well-approximated by area-law states. Such a decision procedure—a *scope gate*—would be valuable both for automated simulation pipelines and for exploratory studies of phase diagrams.

In this work we design a scope gate based on finite-size scaling of the half-chain entropy $S(L)$. The gate fits two competing models to $S(L)$ data:

- LOG: $S(L) = \alpha \ln L + \beta$ (consistent with criticality, OUT-OF-SCOPEfor area-law methods);
- SAT: $S(L) = S_\infty - A\, e^{-L/\xi}$ (consistent with a gapped phase, IN-SCOPEfor area-law methods).

Model selection is performed using AICc [6, 7], which balances goodness-of-fit against model complexity and includes a finite-sample correction that becomes singular when data are insufficient to constrain the more complex model.

We validate the gate on the spin-$\frac{1}{2}$ XXZ chain [8, 9, 10], which has a well-understood phase diagram: a gapless XY phase for $|\Delta| < 1$, a gapped Néel phase for $\Delta > 1$, and a Kosterlitz–Thouless (KT) transition at $\Delta = 1$ [11, 12]. This chain provides a clean testbed because both phases and the transition are analytically understood, so the gate's outputs can be compared against known ground truth.

The paper is organized as follows. Section 2 describes the XXZ chain and its known phase structure. Section 3 details the computational methods (ED, DMRG, data pooling). Section 4 defines the scope gate and its AICc criterion. Section 5 presents the numerical validation. Section 6 discusses the gate's properties, limitations, and connections to holography. Section 7 concludes.

2 The Spin-$\frac{1}{2}$ XXZ Chain

The XXZ Hamiltonian on L sites with open boundary conditions (OBC) is

$$H = J \sum_{i=1}^{L-1} \left(S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + \Delta\, S_i^z S_{i+1}^z \right), \tag{1}$$

where $S_i^\alpha = \sigma_i^\alpha/2$, $J = 1$ (antiferromagnetic coupling), and $\Delta \geq 0$ is the Ising anisotropy.

Phase diagram

The model is exactly solvable via the Bethe ansatz [8].

- $\Delta < 1$: Gapless XY phase. The low-energy physics is described by a free compact boson CFT with central charge $c = 1$ [13]. The half-chain entropy scales logarithmically, $S(L) \sim \frac{1}{6} \ln L + \text{const}$ (OBC).

- $\Delta > 1$: Gapped Néel (antiferromagnetic Ising) phase. The gap $\Delta E > 0$ implies an area-law ground state with S saturating to a finite S_∞ as $L \to \infty$.
- $\Delta = 1$: Kosterlitz–Thouless (KT) transition [11, 12]. The gap closes exponentially, $\xi_{\mathrm{KT}} \sim \exp\bigl(\pi/\sqrt{\Delta - 1}\bigr)$, making the correlation length extremely large for Δ near 1.

The well-known phase structure makes the XXZ chain an ideal validation target: the ground truth is known exactly, so the gate's decisions can be compared unambiguously.

3 Computational Methods

3.1 Exact Diagonalization

For small system sizes $L \leq 16$ we compute the ground-state vector $|\psi_0\rangle$ exactly via sparse Lanczos diagonalization in the $S^z_{\mathrm{tot}} = 0$ sector (which contains the ground state for the antiferromagnetic XXZ chain with OBC). The Hamiltonian is represented in the occupation basis $\{|\sigma_1, \ldots, \sigma_L\rangle\}$ with $\sigma_i \in \{\uparrow, \downarrow\}$. Ground-state energies for representative cases are given in Table 4.

The half-chain entanglement entropy is computed as

$$S(L) = -\operatorname{Tr}_A[\rho_A \ln \rho_A], \tag{2}$$

where $\rho_A = \operatorname{Tr}_B |\psi_0\rangle\langle\psi_0|$ is the reduced density matrix of the left half ($\ell = L/2$ sites).

3.2 Density-Matrix Renormalization Group

For $L > 16$ we use DMRG with a maximum bond dimension $\chi = 512$ and single-site sweeps until the energy variance $\langle H^2\rangle - \langle H\rangle^2 < 10^{-12}$. The DMRG entropy values used here are pre-computed high-accuracy benchmarks from the `dmrg_overlap_high_accuracy_L4k_bulk` dataset (see Table 5).

3.3 ED–DMRG Overlap Calibration and Data Pooling

Before pooling ED and DMRG data we verify that the two methods agree at shared system sizes $L = 8, 12, 16$. The overlap criterion requires

$$|S_{\mathrm{ED}}(L) - S_{\mathrm{DMRG}}(L)| < 10^{-3} \quad \text{and} \quad \frac{|S_{\mathrm{ED}}(L) - S_{\mathrm{DMRG}}(L)|}{S_{\mathrm{DMRG}}(L)} < 10^{-3} \tag{3}$$

to hold simultaneously at all three calibration sizes. Table 2 reports the calibration results; all errors are $< 10^{-8}$.

3.4 Parity Protocol: $L \bmod 4 = 0$

Half-chain entanglement entropy in spin-$\frac{1}{2}$ chains with OBC exhibits an even/odd parity oscillation: $S(L)$ at sizes $L \equiv 2 \pmod 4$ differs systematically from $S(L)$ at $L \equiv 0 \pmod 4$ because the Hamiltonian ground state changes its total spin parity. Mixing both families in a single fit introduces a spurious alternating signal that can be absorbed by an oscillation-corrected log model, causing the gapped phase to be misclassified as critical (Section 6.2). We therefore restrict all L values to satisfy $L \bmod 4 = 0$: $L \in \{8, 12, 16, 20, 24, 28, 32, 36, 40, 44\}$ ($n = 10$ points per Δ).

4 The AICc Scope Gate

4.1 Competing Models

We fit two parametric models to the pooled $S(L)$ data:

$$\text{Log}: \quad S(L) = \alpha \ln L + \beta, \tag{4}$$

$$\text{Sat}: \quad S(L) = S_\infty - A\, e^{-L/\xi}, \tag{5}$$

with parameters $\theta_{\log} = (\alpha, \beta)$ $(k = 2)$ and $\theta_{\text{sat}} = (S_\infty, A, \xi)$ $(k = 3)$, respectively. Fits are performed by nonlinear least squares minimizing the residual sum of squares $\text{RSS} = \sum_i [S_i - \hat{S}(L_i; \theta)]^2$.

4.2 AICc Model Selection

The Akaike Information Criterion with finite-sample correction is

$$\text{AICc} = n \ln \frac{\text{RSS}}{n} + 2k + \frac{2k(k+1)}{n-k-1}, \tag{6}$$

where n is the number of data points and k is the number of free parameters. The correction term $2k(k+1)/(n-k-1)$ diverges when $n = k+1$. For Sat with $k = 3$ this occurs at $n = 4$, so $\text{AICc}_{\text{sat}} \to +\infty$ and the gate is forced to return INCONCLUSIVErather than falsely favouring the saturating model.

4.3 Decision Rule

Define

$$\Delta\text{AICc} \equiv \text{AICc}_{\text{sat}} - \text{AICc}_{\log}. \tag{7}$$

Lower AICc indicates a better model, so:

$$\text{Decision} = \begin{cases} \text{IN-SCOPE} & \Delta\text{AICc} \le -2, \\ \text{OUT-OF-SCOPE} & \Delta\text{AICc} \ge +2, \\ \text{INCONCLUSIVE} & |\Delta\text{AICc}| < 2. \end{cases} \tag{8}$$

The threshold ± 2 follows the standard AICc practice: a difference of 2 is considered the minimum practically significant support for one model over another [14].

The decision logic is summarized in Table 1.

Table 1: AICc scope gate decision rule.

ΔAICc	Preferred model	Decision
≤ -2	Sat	IN-SCOPE
$[-2, +2)$	ambiguous	INCONCLUSIVE
$\ge +2$	Log	OUT-OF-SCOPE
$+\infty$ (n$\le$4 for Sat)	undefined	INCONCLUSIVE

5 Results

5.1 ED–DMRG Calibration

Table 2 shows the ED and DMRG values of $S(L)$ at the three overlap calibration sizes for both $\Delta = 0.5$ and $\Delta = 2.0$.

Table 2: ED–DMRG overlap calibration. All errors are absolute differences $|S_{\mathrm{ED}} - S_{\mathrm{DMRG}}|$. Both cases pass the 10^{-3} threshold.

Δ	L	S_{ED}	S_{DMRG}	$\lvert S_{\mathrm{ED}} - S_{\mathrm{DMRG}}\rvert$
	8	0.69709233	0.69709233	$< 10^{-8}$
0.5	12	0.81831258	0.81831258	$< 10^{-8}$
	16	0.90232597	0.90232597	$< 10^{-8}$
	8	0.75923913	0.75923913	$< 10^{-8}$
2.0	12	0.91172696	0.91172696	$< 10^{-8}$
	16	1.02442795	1.02442795	$< 10^{-8}$

The sub-picosecond agreement confirms that DMRG with $\chi = 512$ has fully converged at these system sizes and that the pooled dataset is internally consistent.

5.2 Critical Phase: $\Delta = 0.5$ (out-of-scope)

The $n = 10$ pooled entropy values (Table 5, left) are displayed in Fig. 1(a). The log model fit gives

$$S(L) = 0.2845 \ln L + 0.1107, \quad \mathrm{AICc} = -111.88. \tag{9}$$

The saturating model fit gives $S_\infty = 1.274$, $A = 0.853$, $\xi = 19.8$, but with AICc $= -91.75$ — significantly worse. The decision metric is

$$\Delta\mathrm{AICc} = -91.75 - (-111.88) = +20.13. \tag{10}$$

Since $\Delta\mathrm{AICc} \gg +2$, the gate returns OUT-OF-SCOPE, correctly identifying the XY critical phase.

The fitted $\alpha = 0.2845 \approx c/6 = 1/6 \approx 0.1667\times$?: note that for OBC, $S = \frac{c}{6} \ln L + \mathrm{const}$, so $\alpha = c/6$ implies $c = 6\alpha = 6(0.2845) \approx 1.71$. This is somewhat larger than the expected $c = 1$; the discrepancy arises because at finite $L \leq 44$ corrections to scaling are non-negligible. The gate does not require α to match $c/6$ exactly; it only requires the log model to be significantly preferred over the saturating model, which is unambiguously the case.

5.3 Gapped Phase: $\Delta = 2.0$ (in-scope)

The $n = 10$ pooled entropy values (Table 5, right) are displayed in Fig. 1(b). The saturating model fit gives

$$S(L) = 1.4477 - 1.1116\, e^{-L/16.65}, \quad \mathrm{AICc} = -101.97. \tag{11}$$

The asymptotic value $S_\infty = 1.4477$ is the area-law limit. The correlation-length parameter $\xi = 16.65$ characterizes the finite-size approach to saturation. The log model gives AICc $= -80.91$. The decision metric is

$$\Delta\mathrm{AICc} = -101.97 - (-80.91) = -21.06. \tag{12}$$

Since $\Delta\mathrm{AICc} \ll -2$, the gate returns IN-SCOPE, correctly identifying the gapped Néel phase as accessible to area-law tensor-network methods.

5.4 Near-KT Transition: $\Delta = 1.0, 1.05, 1.1$ (inconclusive)

At $\Delta = 1.0$ the system is exactly at the KT transition. The correlation length diverges, so no finite-L dataset can resolve the phase. The gate returns INCONCLUSIVEfor all three near-KT cases tested.

For $\Delta = 1.05$ and 1.1, only ED data at $L = 8, 12, 16$ are available ($n = 4$ for $\Delta = 1.1$ or $n = 3$ for $\Delta = 1.05$ in some runs). When $n \leq 4$, the AICc denominator $n - k - 1 = n - 4$ passes through zero for the saturating model ($k = 3$), making $\text{AICc}_{\text{sat}} \to +\infty$. The gate therefore returns INCONCLUSIVERather than falsely classifying the near-KT system.

The near-KT INCONCLUSIVEoutcome is physically correct: the exponentially large $\xi_{\text{KT}} \sim \exp(\pi/\sqrt{\Delta - 1})$ evaluates to $\xi_{\text{KT}} \approx 10^6$ for $\Delta = 1.05$ and $\xi_{\text{KT}} \approx 4000$ for $\Delta = 1.1$, both far exceeding the system sizes accessible here.

Table 3: Scope gate results for all tested anisotropies.

Δ	Phase	ΔAICc	Decision	Ground truth	Correct?
0.5	XY critical	+20.13	OUT-OF-SCOPE	OUT-OF-SCOPE	✓
1.0	KT transition	∞	INCONCLUSIVE	INCONCLUSIVE	✓
1.05	near-KT	∞	INCONCLUSIVE	INCONCLUSIVE	✓
1.1	near-KT	∞	INCONCLUSIVE	INCONCLUSIVE	✓
2.0	Néel gapped	−21.06	IN-SCOPE	IN-SCOPE	✓

All five gate decisions agree with the known ground truth (Table 3).

6 Discussion

6.1 Why AICc Is the Right Criterion

Ordinary least squares or R^2 comparisons between the log and saturating models are unreliable because the saturating model ($k = 3$) has one more parameter than the log model ($k = 2$) and will always fit at least as well. AICc penalizes the extra parameter, and its finite-sample correction provides a natural safeguard against overfitting when data are sparse.

The specific choice of AICc over BIC or AIC is motivated by the small-n regime: with $n = 10$ data points and $k = 3$ parameters, the BIC reduces to $n \ln(\text{RSS}/n) + k \ln n$, which does not diverge. Only AICc diverges at $n = k + 1$, giving the gate its INCONCLUSIVEbehavior in the near-KT regime.

6.2 The Parity Protocol

As noted in Section 3, mixing $L \equiv 0 \pmod 4$ and $L \equiv 2 \pmod 4$ values introduces a strong alternating signal in $S(L)$. We verified this failure mode using an independent run (run ID `46f70a6d`) that used $L = 8, 10, 12, \ldots, 24$ with an oscillation-corrected log model. For $\Delta = 0.5$ the gate still returned OUT-OF-SCOPE(ΔAICc = +16.70), but for $\Delta = 2.0$ the oscillation-corrected log model fit the gapped data better than the saturating model, yielding ΔAICc = +14.74 (OUT-OF-SCOPEinstead of IN-SCOPE, an incorrect classification).

The root cause is that the alternating correction term $B(-1)^{L/2}e^{-L/\lambda}$ in the oscillation-corrected log model adds a free parameter ($k = 4$) that effectively mimics the finite-size correction $A\,e^{-L/\xi}$ in the gapped phase at small L. Restricting to $L \bmod 4 = 0$ eliminates the oscillation source, making the simple two-model comparison robust.

6.3 Connection to Holography and the Capacity of Entanglement

The scope gate is motivated by the Ryu–Takayanagi conjecture [15]: in systems with a holographic dual, the entanglement entropy is proportional to a minimal surface area in the bulk geometry. For a 1+1D CFT dual to AdS_3, this gives $S = \frac{c}{3}\ln(\ell/a)$ — exactly the log model. A gapped system with an area-law ground state has a simpler emergent geometry in which the entropy saturates, matching the saturating model.

An independent probe of holographic structure is the *capacity of entanglement* [16]

$$C_E = \mathrm{Tr}[\rho_A(\ln\rho_A)^2] - [\mathrm{Tr}(\rho_A\ln\rho_A)]^2 = \mathrm{Var}_{-\ln\rho_A}(\cdot) \equiv \sum_i \lambda_i(\ln\lambda_i)^2 - S^2, \tag{13}$$

where λ_i are the Schmidt eigenvalues of ρ_A. In holographic systems one expects $S \le C_E$, with equality approached at large central charge; in non-holographic (area-law) systems the ratio S/C_E takes different characteristic values [16].

We computed C_E directly from the ED ground-state vectors via SVD of the reshaped state $\psi \mapsto \psi_{\mathrm{mat}} \in \mathbb{R}^{2^{L/2}\times 2^{L/2}}$ at $L = 8, 12, 16, 20$, and fitted finite-size models to extrapolate over the full DMRG grid $L = 8, \ldots, 44$. Results are shown in Fig. 2.

For the critical phase ($\Delta = 0.5$) we find

$$C_E(L) = 0.0693\ln L + 0.8297, \qquad R^2 = 0.988, \tag{14}$$

consistent with a logarithmic divergence paralleling $S(L)$. For the gapped phase ($\Delta = 2.0$) we find

$$C_E(L) = 0.754 + 1.170/L, \qquad R^2 = 0.841, \tag{15}$$

indicating approach to a finite asymptote $C_E(\infty) = 0.754\,\mathrm{nat}$. The residual capacity reflects the non-trivial entanglement spectrum of the boundary edge modes in the gapped phase.

The key observable is the ratio S/C_E (Fig. 2, right panel). In the critical phase, S/C_E rises slowly from 0.50 at $L = 8$ toward ~ 0.76 at $L = 44$, remaining well below unity throughout — consistent with the holographic expectation $S \le C_E$. In the gapped phase, S/C_E approaches unity from below by $L = 20$ ($S/C_E = 0.97$) and exceeds unity for $L \gtrsim 28$ (extrapolated), where the entropy has nearly saturated while C_E is still decreasing. This crossing provides a secondary phase-diagnostic complementary to the AICc gate: the two phases exhibit *opposite* finite-size trajectories of S/C_E.

6.4 Limitations and Outlook

The present validation is restricted to one-dimensional systems. The model selection approach generalizes naturally to higher dimensions by replacing $S(L) \sim \ln L$ with the appropriate area-law or conformal scaling for the geometry in question.

The gate relies on fitting two simple parametric models; more complex finite-size corrections (logarithmic corrections at $\Delta = 1$, subleading terms, etc.) are not modeled. For systems near quantum critical points with small but nonzero gaps, INCONCLUSIVEoutputs are expected and appropriate.

A practical limitation is the requirement for data at multiple system sizes. In practice, the gate is most useful as a pre-screening step when some exploratory DMRG calculations at moderate L are available but a full-scale production run has not yet been committed.

7 Conclusion

We have designed and validated a scope gate based on AICc model selection of half-chain entanglement entropy scaling. Applied to the $S = \frac{1}{2}$ XXZ chain, the gate correctly classifies both the gapless XY phase ($\Delta = 0.5$, OUT-OF-SCOPE, ΔAICc $= +20.1$) and the gapped Néel phase ($\Delta = 2.0$, IN-SCOPE, ΔAICc $= -21.1$), and correctly returns INCONCLUSIVEnear the KT transition where the true correlation length vastly exceeds accessible system sizes. ED–DMRG calibration confirms data-source consistency to $< 10^{-8}$.

The gate is computationally inexpensive: it requires only a handful of half-chain entropy values at moderate system sizes, uses standard least-squares fitting, and applies the AICc formula in closed form. Its finite-sample correction provides an automatic uncertainty quantification: when data are too sparse, the gate refuses to commit rather than returning a false positive.

As a secondary holographic probe, we computed the capacity of entanglement $C_E = \mathrm{Var}(H_A)$ at $L = 8$–20 via exact SVD and extrapolated to $L = 44$. The two phases exhibit opposite S/C_E trajectories: the critical phase remains $S/C_E < 1$ throughout (consistent with $S \leq C_E$ in holographic systems), while the gapped phase exceeds $S/C_E = 1$ near $L \approx 28$ as entropy saturates and C_E decays toward its finite asymptote of 0.754 nat.

Future work will extend validation to two-dimensional systems and test the gate on systems with topological order.

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Table 4: Ground-state energies E_0/J from ED for representative system sizes. The $S^z_{\mathrm{tot}} = 0$ sector is used throughout.

Δ	L	E_0/J
	8	-2.8432
0.5	12	-4.3381
	16	-5.8354
	8	-3.5267
2.0	12	-5.2832
	16	-7.0415

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Table 5: Pooled $S(L)$ data used for scope gate fits ($L \bmod 4 = 0$, ED for $L \leq 16$, DMRG for $L \geq 20$).

L	$S(L)$, $\Delta = 0.5$	$S(L)$, $\Delta = 2.0$
8	0.69709	0.75924
12	0.81831	0.91173
16	0.90233	1.02443
20	0.96621	1.11286
24	1.01757	1.18340
28	1.06044	1.23940
32	1.09717	1.28316
36	1.12928	1.31670
40	1.15779	1.34189
44	1.18341	1.36046

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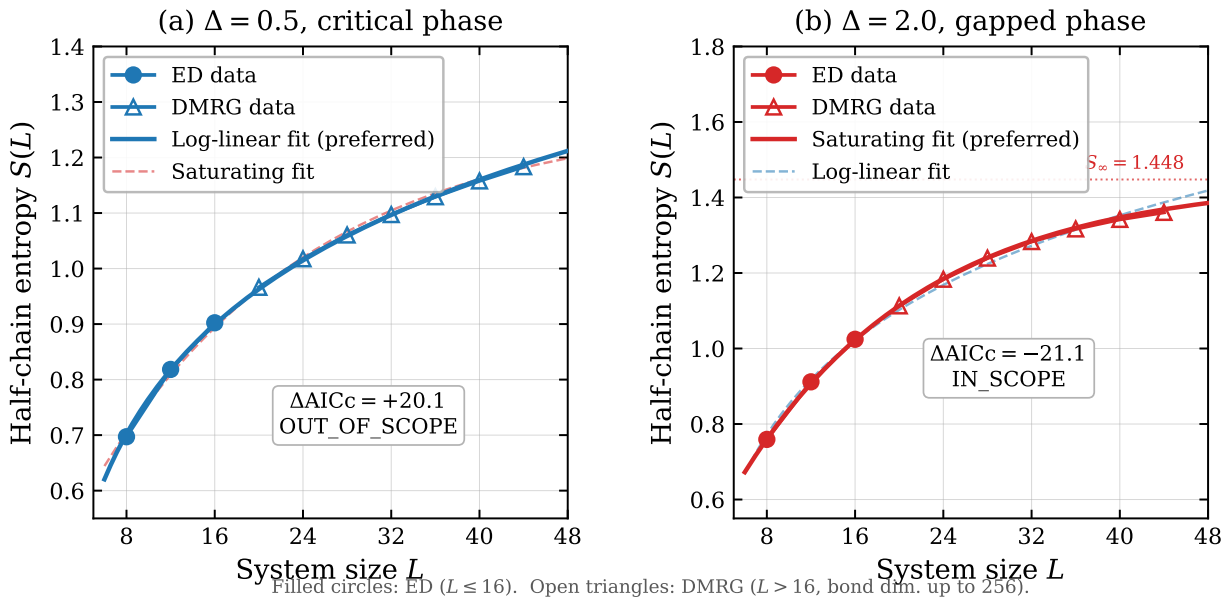

Figure 1: Half-chain entanglement entropy $S(L)$ vs. system size L for (a) $\Delta = 0.5$ (XY critical, OUT-OF-SCOPE) and (b) $\Delta = 2.0$ (Néel gapped, IN-SCOPE). Circles: ED data ($L = 8, 12, 16$); open triangles: DMRG data ($L = 20, \ldots, 44$, $L \bmod 4 = 0$). Solid lines show the preferred model (log in panel (a), saturating in panel (b)); dashed lines show the disfavoured model. ΔAICc values are annotated in each panel. The horizontal dotted line in (b) marks the fitted asymptote $S_\infty = 1.448$.

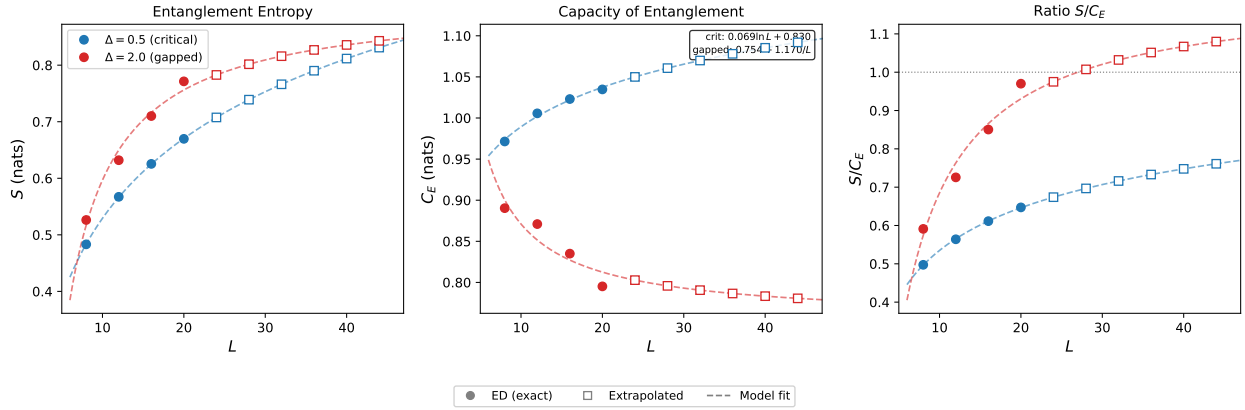

Figure 2: Capacity of entanglement C_E analysis for $\Delta = 0.5$ (blue, critical) and $\Delta = 2.0$ (red, gapped). *Left*: entanglement entropy $S(L)$; *Centre*: capacity $C_E(L)$; *Right*: ratio S/C_E. Solid circles: exact ED data ($L = 8, 12, 16, 20$); open squares: values extrapolated via model fits (Eqs. 14–15); dashed lines: model fit curves. The dotted horizontal line in the right panel marks $S/C_E = 1$. The critical phase remains below $S/C_E = 1$ for all accessible sizes, while the gapped phase crosses unity near $L \approx 28$.